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/* =====
RaceDay - Database Schema and Seed Data
Target: SQL Server (SSMS)
Matches the ERD in /docs/RaceDay_ERD.png
===== */

IF DB_ID('RaceDayDB') IS NULL
BEGIN
    CREATE DATABASE RaceDayDB;
END
GO

USE RaceDayDB;
GO

/* ----- Drop tables if they already exist (clean rebuild) ----- */
IF OBJECT_ID('dbo.Results', 'U') IS NOT NULL DROP TABLE dbo.Results;
IF OBJECT_ID('dbo.Enrolments', 'U') IS NOT NULL DROP TABLE dbo.Enrolments;
IF OBJECT_ID('dbo.RouteInfo', 'U') IS NOT NULL DROP TABLE dbo.RouteInfo;
IF OBJECT_ID('dbo.Categories', 'U') IS NOT NULL DROP TABLE dbo.Categories;
IF OBJECT_ID('dbo.Events', 'U') IS NOT NULL DROP TABLE dbo.Events;
IF OBJECT_ID('dbo.Users', 'U') IS NOT NULL DROP TABLE dbo.Users;
GO

/* =====
TABLE: Users
===== */
CREATE TABLE dbo.Users (
    UserID          INT IDENTITY(1,1) PRIMARY KEY,
    FullName        NVARCHAR(100)    NOT NULL,
    Email           NVARCHAR(150)    NOT NULL UNIQUE,
    PasswordHash    NVARCHAR(255)    NOT NULL,
    Role            NVARCHAR(20)     NOT NULL CHECK (Role IN ('Organiser', 'Partic
    Phone           NVARCHAR(20)     NULL,
    CreatedAt       DATETIME2        NOT NULL DEFAULT (SYSDATETIME())
);
GO

/* =====
TABLE: Events
===== */
CREATE TABLE dbo.Events (
    EventID         INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID     INT              NOT NULL,

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        Name                NVARCHAR(150)    NOT NULL,
        Description          NVARCHAR(1000)   NULL,
        EventDate            DATETIME2        NOT NULL,
        Province             NVARCHAR(50)     NOT NULL,
        Venue                NVARCHAR(150)    NOT NULL,
        EventType            NVARCHAR(20)     NOT NULL CHECK (EventType IN ('Run', 'Walk',
        CreatedAt            DATETIME2        NOT NULL DEFAULT (SYSDATETIME()),
        CONSTRAINT FK_Events_Organiser FOREIGN KEY (OrganiserID) REFERENCES dbo.Users
    );
GO

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/* =====
TABLE: Categories
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CREATE TABLE dbo.Categories (
    CategoryID      INT IDENTITY(1,1) PRIMARY KEY,
    EventID         INT                NOT NULL,
    Name            NVARCHAR(50)       NOT NULL,
    DistanceKm      DECIMAL(5,2)       NOT NULL,
    MaxParticipants INT                NOT NULL DEFAULT (500),
    EntryFee        DECIMAL(8,2)       NOT NULL DEFAULT (0.00),
    CONSTRAINT FK_Categories_Events FOREIGN KEY (EventID) REFERENCES dbo.Events(E
);
GO

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/* =====
TABLE: RouteInfo (one route per category)
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CREATE TABLE dbo.RouteInfo (
    RouteID         INT IDENTITY(1,1) PRIMARY KEY,
    CategoryID      INT                NOT NULL UNIQUE,
    RouteMapURL     NVARCHAR(255)      NULL,
    StartPoint      NVARCHAR(150)      NOT NULL,
    EndPoint        NVARCHAR(150)      NOT NULL,
    ElevationGainM  INT                NULL DEFAULT (0),
    Description      NVARCHAR(1000)    NULL,
    CONSTRAINT FK_RouteInfo_Categories FOREIGN KEY (CategoryID) REFERENCES dbo.Ca
);
GO

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/* =====
TABLE: Enrolments
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CREATE TABLE dbo.Enrolments (
    EnrolmentID     INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID   INT                NOT NULL,
    CategoryID      INT                NOT NULL,

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        EnrolmentDate    DATETIME2          NOT NULL DEFAULT (SYSDATETIME()),
        RaceNumber       NVARCHAR(10)       NULL,
        Status           NVARCHAR(20)       NOT NULL DEFAULT ('Pending') CHECK (Status IN
CONSTRAINT FK_Enrolments_Users FOREIGN KEY (ParticipantID) REFERENCES dbo.Use
CONSTRAINT FK_Enrolments_Categories FOREIGN KEY (CategoryID) REFERENCES dbo.C
CONSTRAINT UQ_Enrolment_Once UNIQUE (ParticipantID, CategoryID)
);
GO

/* =====
TABLE: Results (one result per enrolment)
===== */
CREATE TABLE dbo.Results (
    ResultID              INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID           INT                NOT NULL UNIQUE,
    FinishTime            TIME                NULL,
    Position              INT                NULL,
    Status                NVARCHAR(20)       NOT NULL DEFAULT ('Finished') CHECK (
    CapturedByOrganiserID INT                NOT NULL,
    CapturedAt            DATETIME2          NOT NULL DEFAULT (SYSDATETIME()),
    CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentID) REFERENCES dbo.Enr
    CONSTRAINT FK_Results_Organiser FOREIGN KEY (CapturedByOrganiserID) REFERENCE
);
GO

/* =====
SEED DATA
===== */

-- Organisers (2) and Participants (2)
INSERT INTO dbo.Users (FullName, Email, PasswordHash, Role, Phone) VALUES
('Thandiwe Mokoena', 'thandiwe@raceday.co.za', 'HASHED_PWD_1', 'Organiser', '0821
('Johan van der Merwe', 'johan@raceday.co.za', 'HASHED_PWD_2', 'Organiser', '0837
('Lerato Dlamini', 'lerato@example.com', 'HASHED_PWD_3', 'Participant', '07311122
('Sipho Nkosi', 'sipho@example.com', 'HASHED_PWD_4', 'Participant', '0729998888')
GO

-- Events (3)
INSERT INTO dbo.Events (OrganiserID, Name, Description, EventDate, Province, Venu
(1, 'Comrades Marathon', 'Iconic ultramarathon between Pietermaritzburg and Durba
(1, 'Cape Town Cycle Tour', 'Scenic cycling tour around the Cape Peninsula.', '20
(2, 'Soweto Community Fun Walk', 'Community charity walk supporting local schools
GO

-- Categories (across the 3 events)
INSERT INTO dbo.Categories (EventID, Name, DistanceKm, MaxParticipants, EntryFee)
(1, '87km Ultra', 87.00, 20000, 950.00),

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(1, '10km Fun Run', 10.00, 5000, 250.00),
(2, '109km Cycle Race', 109.00, 15000, 750.00),
(2, '35km Mini Tour', 35.00, 6000, 400.00),
(3, '5km Fun Walk', 5.00, 2000, 100.00);
GO
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-- Route info for a couple of categories
INSERT INTO dbo.RouteInfo (CategoryID, RouteMapURL, StartPoint, EndPoint, Elevati
(1, 'https://maps.example.com/comrades-87km', 'Pietermaritzburg City Hall', 'King
(3, 'https://maps.example.com/cycle-tour-109km', 'Green Point Common', 'Green Poi
GO
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-- Enrolments (participants entering categories)
INSERT INTO dbo.Enrolments (ParticipantID, CategoryID, RaceNumber, Status) VALUES
(3, 1, 'A1023', 'Confirmed'),
(4, 3, 'B2041', 'Confirmed'),
(3, 5, 'C3070', 'Pending');
GO
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-- Sample results captured by an organiser
INSERT INTO dbo.Results (EnrolmentID, FinishTime, Position, Status, CapturedByOrg
(1, '08:32:15', 4521, 'Finished', 1),
(2, '03:45:02', 812, 'Finished', 1);
GO
```